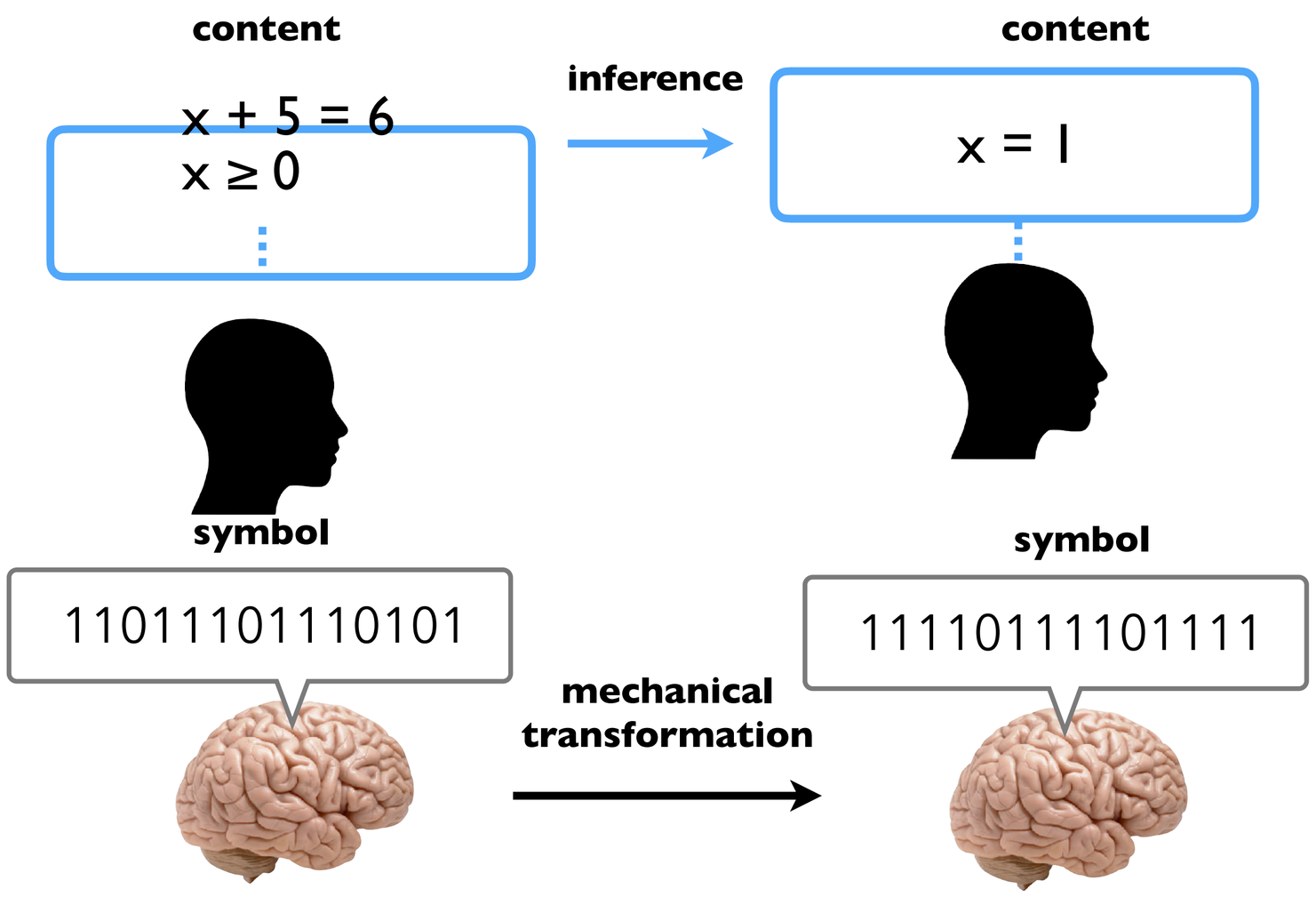
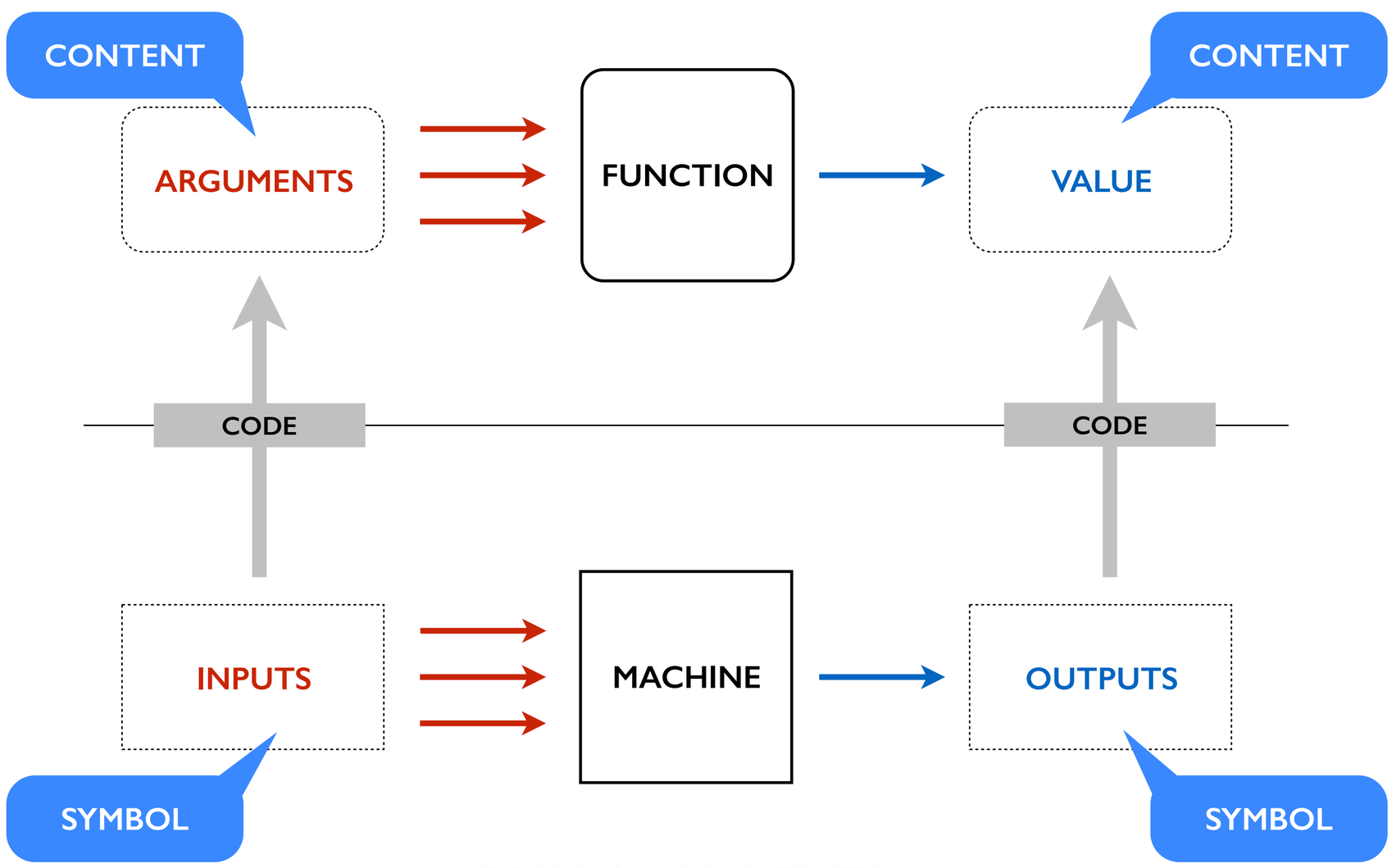
40. Computational Theory of Mind: The Long View

Cognition as Computation

> My hypothesis then is that thought models, or parallels, reality—that its essential feature is not 'the mind', 'the self', 'sense-data', nor propositions but symbolism, and that this symbolism is largely of the same kind as that which is familiar to us in mechanical devices which aid thought and calculation.

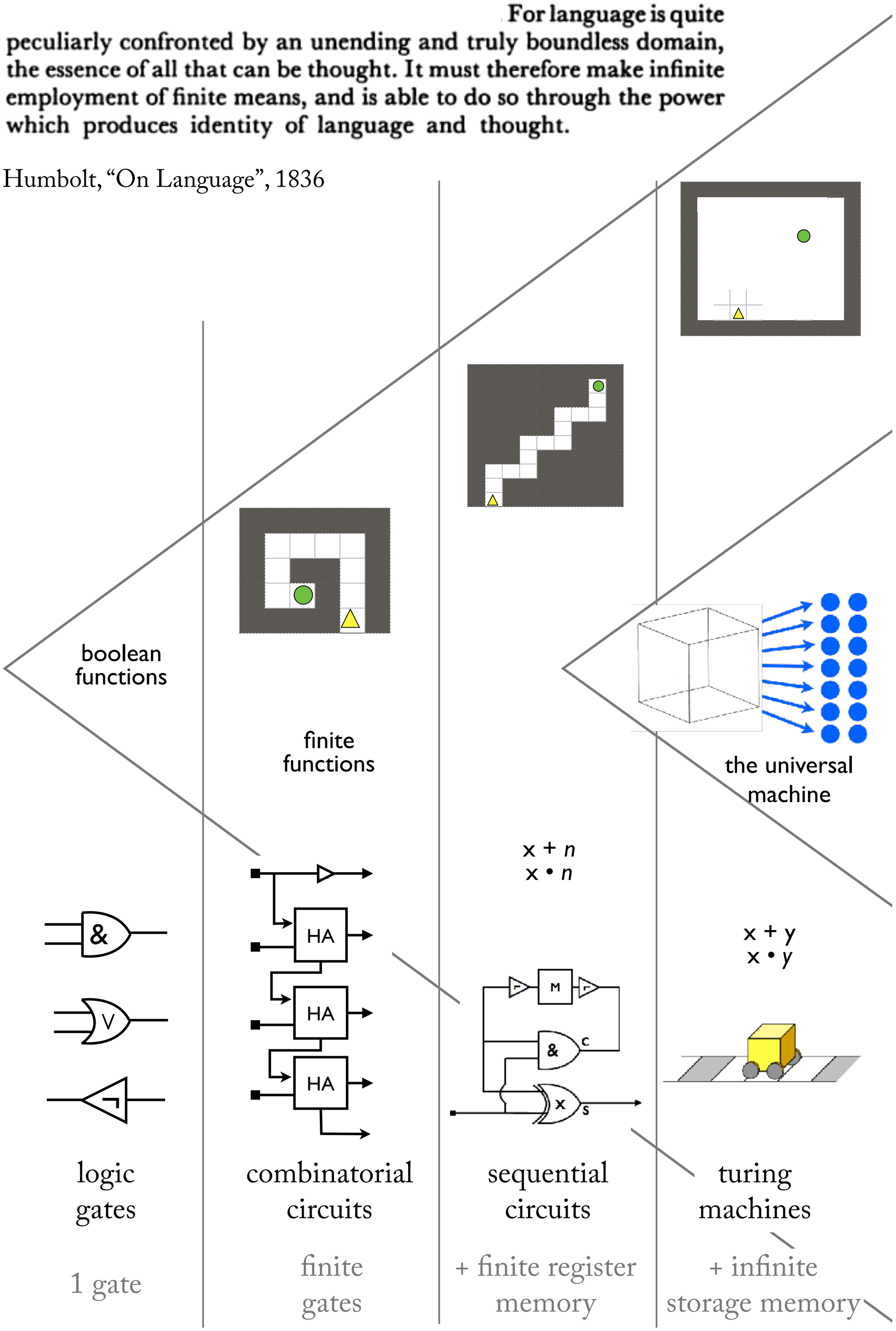




Orders of Computational Power

> For language is quite peculiarly confronted by an unending and truly boundless domain, the essence of all that can be thought. It must therefore make infinite employment of finite means, and is able to do so through the power which produces identity of language and thought.

Humboldt, "On Language", 1836



Levels of Computational Abstraction

"If one hopes to achieve a full understanding of a system as complicated as a nervous system… then one must be prepared to contemplate different kinds of explanation at different levels of description that are linked, at least in principle, into a cohesive whole, even if linking the levels in complete detail is impractical."

Marr, *Vision*, 1982

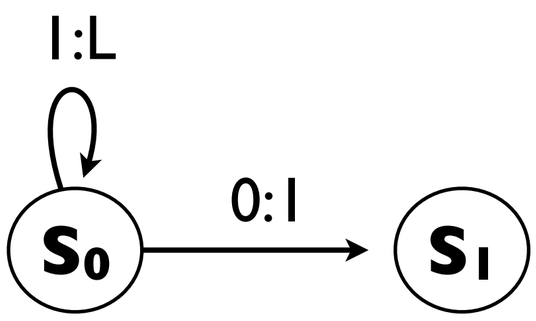
Ecological-Cognitive Level

+1

Functions

Semantic Abstraction

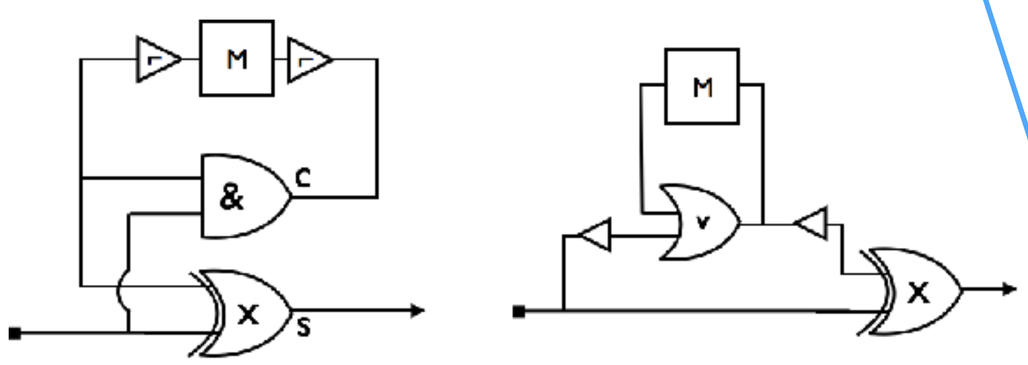
Algorithmic Level



State Machines

State Abstraction

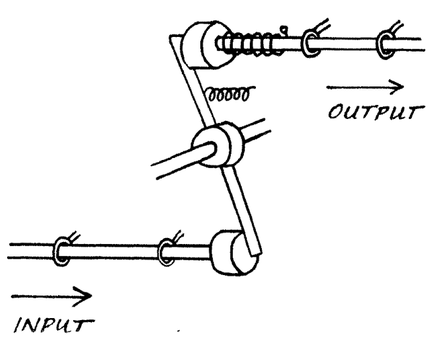
Mechanical Level

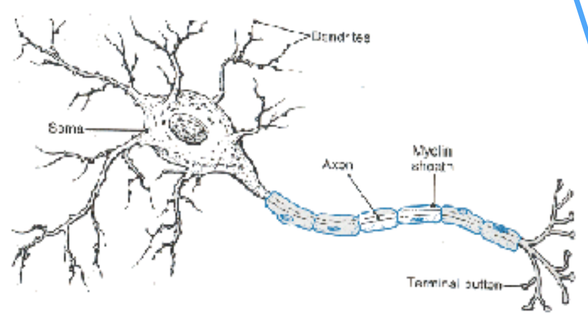


Logic Circuits

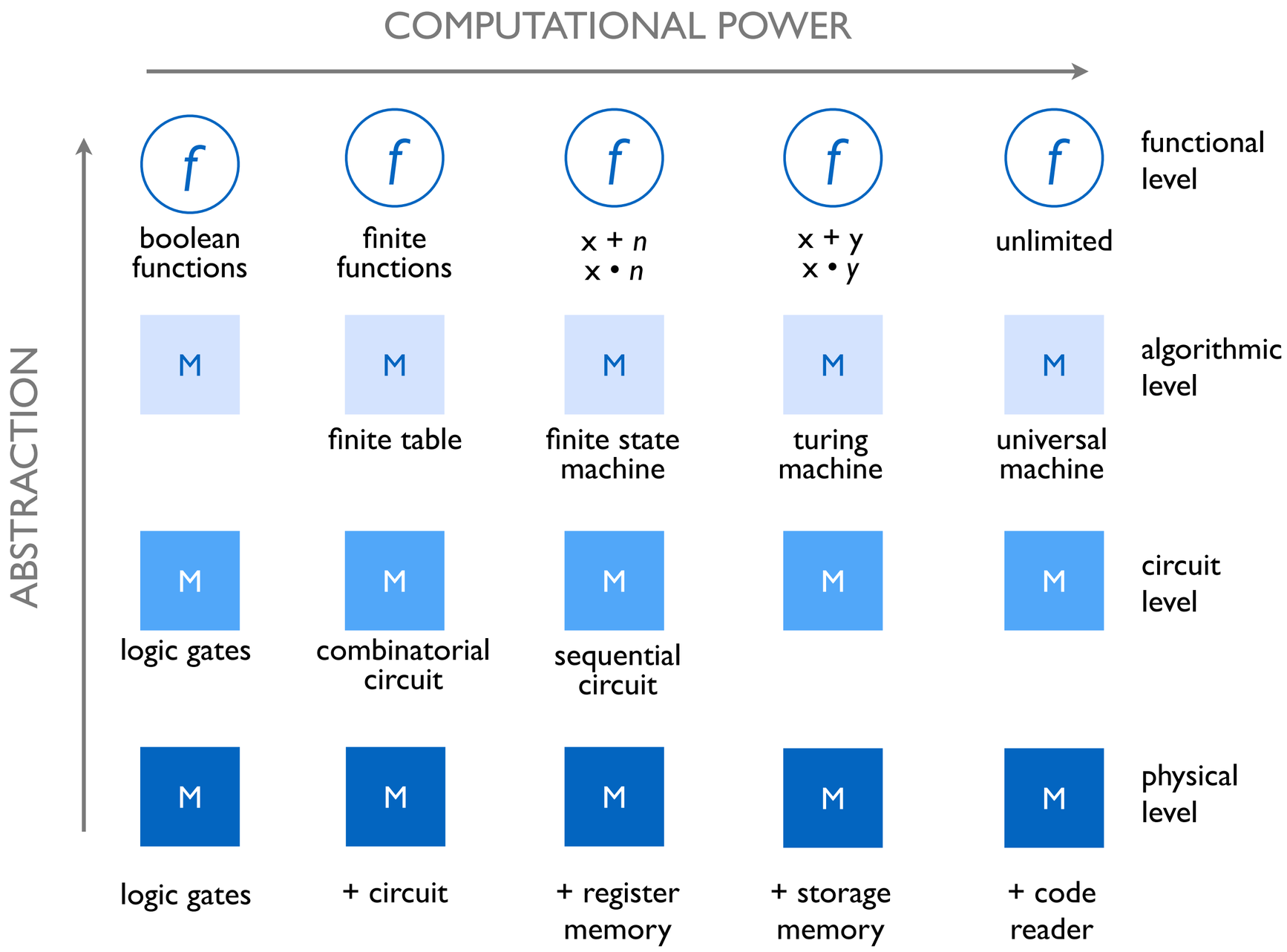
Functional Abstraction

Physical Level

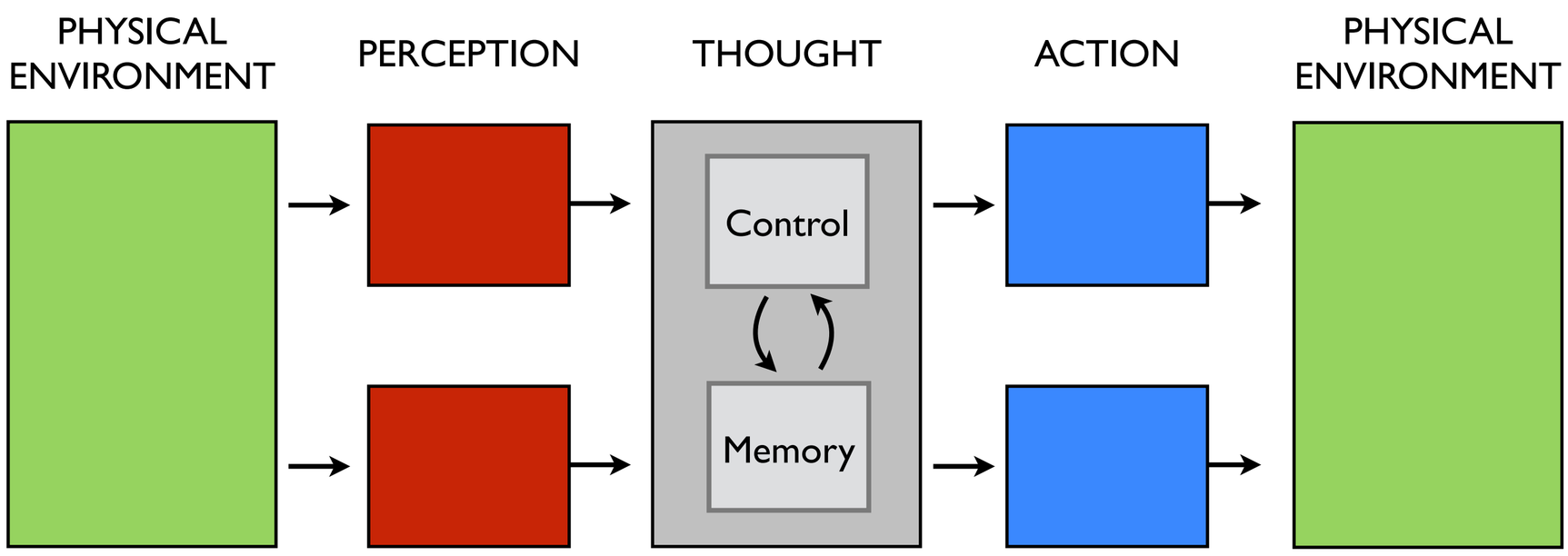


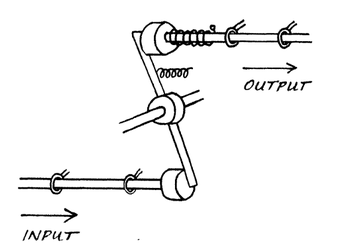


Physical Machines



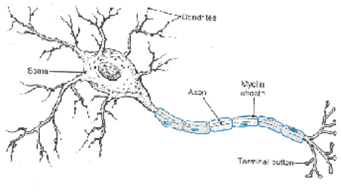
Cognition as Natural Computation







Home is 389 steps away.





I think therefore I am.

Editorial Assistance: Bill Kowalsky, Jon Ma, Christian DeLeon